King Saud University

Collage of Computer Science

Computer Science Department

CSC261 – Project 1

In this project, you will apply machine learning regression techniques to solve two realistic business problems in the used car market. You will work with the provided dataset (Car Dataset) and demonstrate how data can be used to estimate selling prices and predict vehicle characteristics.

This project is designed to help you move from theory to practice: you will explore the data, prepare it, train models, evaluate them, and interpret the results in clear business language.

Project Objectives

By the end of this project, you should be able to:

1. Explain and apply **Linear Regression** to predict a continuous quantity (car selling price).
2. Explain and apply **Logistic Regression** to classify a binary outcome (automatic vs manual transmission).
3. Perform **Exploratory Data Analysis (EDA)** and interpret summary statistics and correlations.
4. Preprocess real data: engineer new features, handle categorical variables, and split data correctly.
5. Evaluate model performance using appropriate metrics.
6. Interpret model outputs (coefficients, error metrics, etc.) in practical/business terms.
7. Improve model quality and reliability using **K-Fold Cross-Validation** if applicable.

You must describe these objectives **in your own words** in your report.

Theoretical Background

In this section of your report, you must explain:

1. Linear Regression

- What problem it solves.
- The idea of fitting a line (or hyperplane) to minimize the sum of squared errors.
- The meaning of model coefficients.

- What it means to “generalize” to unseen data.

2. Logistic Regression

- The general concept of logistic regression.
- Why we cannot use linear regression for classification.
- The logistic (sigmoid) function and how it outputs a probability between 0 and 1.

You are encouraged to include formulas, plots, or diagrams to support your explanation.

Dataset Description

In this section, you must describe the provided dataset (Car Dataset). At a minimum:

- List and describe each feature/column (for example: Year, Selling_Price, Present_Price, Kms_Driven, Fuel_Type, Seller_Type, Transmission, Owner, etc.).
- State which features are **numerical** and which are **categorical**.
- Explain which column will be the **target variable** in Part 1 and which will be the **target variable** in Part 2.
- Mention any new engineered features you create (for example, vehicle age).

You should also explain why some columns may be dropped if needed.

Part 1: Predicting Used Car Selling Price Using Linear Regression

1. Problem Statement

Write briefly, in your own words a clear problem statement. What you are trying to solve.

2. Exploratory Data Analysis (EDA)

Perform and present the following:

- Show the first rows.
- Show dataset shape (number of rows and columns).
- Show dataset info: data types and non-null counts.
- Show descriptive statistics: count, mean, std, min, max, quartiles for numerical columns.
- Create and interpret at least one visualization (for example, a histogram of Selling_Price or Present_Price, or a scatter plot of Present_Price vs Selling_Price).
- Plot a **correlation heatmap** among numerical features and briefly comment on what is strongly correlated with Selling_Price.

You must write a short interpretation of what you observe (do not just paste screenshots).

3. Dataset Preprocessing

Before training the model, you must:

1. **Feature Engineering**
   - Create a new column for vehicle age. Explain why age matters.
2. **Categorical vs Numerical**
   - Clearly list how many categorical features you are using and how many numerical features. Specify each category.
   - Convert categorical variables to numeric.
3. **Train/Test Split**
   - Split the dataset into training and testing portions:
     - **Test size = 20%**
     - **Random state = 1**
4. **Scaling (if you apply scaling)**
   - If you standardize numeric features, explain why.

4. Model Building (Linear Regression)

- Train a Linear Regression model on the training data to predict Selling_Price.
- Show the learned model coefficients and explain (in words) what the most important features seem to be.

5. Model Evaluation

Compute and report:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

For each metric, explain the meaning in plain language:

6. Results and Interpretation

In this section you should discuss you model's results.

7. Model Improvement

Now try to improve the reliability of your regression model using **K-Fold Cross-Validation**.

1. Use K-Fold (for example, KFold or cross_val_score in scikit-learn with cv=5).

2. Evaluate the model's MAE or RMSE across the 5 folds on the training data.
3. Try three different fold scenarios.
4. Report:
    - The mean cross-validation score.
    - The standard deviation across folds.

This might be helpful due to the following:

- A single train/test split can be "lucky" or "unlucky."
- Cross-validation tells us how stable the model is and how well it is expected to generalize.

To evaluate this improvement step, do the following:

- Compare the single-split test error vs. the cross-validated average error.
- State whether your model appears stable or if its performance changes a lot depending on which data it sees.

8. Conclusion for Part 1

Summarize:

- Can we reasonably predict a car's selling price using these features?
- What are the main limitations of the model? Suggest three extra features that could enhance the performance of the model.

Part 2: Predicting Automatic vs. Manual Transmission Using Logistic Regression

1. Problem Statement

Write briefly, in your own words a clear problem statement. What you are trying to solve.

2. Data Preparation

You will repeat a similar pipeline, but now for classification:

- Select the relevant features.
- Identify the Y. i.e. the desired output.
- Convert categorical variables to numerical.
- Train/test split (again, 80% / 20%).
- Scale numeric features if needed.

3. Model Training (Logistic Regression)

- Train a Logistic Regression classifier.
- Show the learned model coefficients and explain (in words) what the most important features seem to be.

4. Model Evaluation

Compute the following:

- Confusion Matrix
- Accuracy
- Precision and Recall for the “Automatic” class

Explain these in plain English:

- Precision: “When the model predicts Automatic, how often is it correct?”
- Recall: “Out of all actual Automatic cars, how many did we successfully catch?”

5. Interpretation

Discuss:

- Which features make a car more likely to be automatic?
  (Look at the logistic regression coefficients. A positive coefficient means that higher values of that feature increase the chance of being Automatic.)
- Are automatic cars newer? More expensive? Lower mileage?

6. Conclusion for Part 2

Summarize:

- How well can we guess transmission type using the available features?

Final Deliverables

Your submission must include:

1. **Written Report (PDF or Word):**
    - Introduction
    - Project Objectives
    - Theoretical Background (Linear vs Logistic Regression)
    - Dataset Description
    - Part 1 (all required subsections, including K-Fold model improvement)
    - Part 2 (all required subsections)
    - Final Conclusions
2. **Code (Jupyter Notebook or .py files):**
    - Clean, runnable code for Part 1 and Part 2.
    - Clearly labeled steps.
    - All plots generated by your code.